Trianz Internal Leadership Dashboard

Project Documentation

Comprehensive Technical & Functional Reference

Prepared by: Deepak V | Trianz Digital Consulting

Version 1.0 | August 2026

Table of Contents

Table of Contents 2
1. Executive Summary 5
Key Highlights 5
2. Project Overview & Scope 6
2.1 Problem Statement 6
2.2 Solution 6
2.3 Scope 6
2.4 End Users 6
3. Technical Architecture 7
3.1 Tech Stack 7
3.2 System Architecture 7
3.3 Project File Structure 7
4. Dashboard Features — By Tab 9
4.1 Finance Tab 9
KPIs & Metrics 9
Filters & Views 9
How Data Flows 9
4.2 HR & People Tab 9
KPIs & Metrics 10
How Data Flows 10
4.3 Delivery Tab 10
KPIs & Metrics 10
How Data Flows 11
4.4 Sales Tab 11
Pipeline Stage Buckets 11
KPIs & Metrics 11
How Data Flows 11
4.5 Active Engagements Tab 12
KPIs & Metrics 12
Migration Urgency Classification 12
Needs Attention Panel 12
4.6 Backlog Tab 13
KPIs & Metrics 13

How Data Flows 13
5. Data Sources & Integration 14
5.1 SharePoint as the Single Source of Truth 14
5.2 SharePoint Folder Structure 14
5.3 Microsoft Graph API Authentication 14
5.4 Environment Variables Required 14
5.5 Fallback Mechanism 14
6. ETL Pipeline Explained 16
6.1 How the ETL Pipeline Works 16
6.2 ETL Scripts Overview 16
6.3 Robustness Design 16
6.4 AI Integration in ETL 17
6.5 Scheduled Refresh 17
7. Advanced Features 18
7.1 AI Leadership Summary 18
7.2 Needs Attention Panel 18
7.3 Drill-Down Drawers & Modals 18
7.4 Export Functionality 18
7.5 Migration Urgency Classification 19
7.6 Period & Offering Filters 19
8. Data & Refresh Operations 20
8.1 How the Refresh Button Works 20
8.2 Refresh Status Indicators 20
8.3 Tab-Specific Refresh 20
8.4 Data Sources Per Tab Summary 20
8.5 Refresh Frequency & Scheduling 21
9. Data Dictionary & Schema 22
9.1 Finance Metrics 22
9.2 HR Metrics 22
9.3 Sales & Pipeline Metrics 22
10. User Guide & Walkthrough 24
10.1 Navigating the Dashboard 24
10.2 Using Filters 24
10.3 Interacting with KPI Cards 24
10.4 Using the AI Leadership Summary 24

10.5 Refreshing Data 24
10.6 Downloading Reports 25
11. Integration Points 26
12. Assumptions & Constraints 27
12.1 Data Assumptions 27
12.2 Known Limitations 27
12.3 What Is Not Covered 27
13. Performance & Reliability 28
14. FAQ & Troubleshooting 29
14.1 Frequently Asked Questions 29
14.2 Common Issues & Fixes 29
15. Appendix 30
15.1 ETL File Reference 30
15.2 API Routes Reference 30
15.3 Glossary 30
15.4 Key Contacts 31

1. Executive Summary

The Trianz Internal Leadership Dashboard is a centralized, real-time business intelligence platform built for Trianz leadership to monitor organizational performance across Finance, Sales, HR & People, Delivery, Active Engagements, and Backlog in a single unified interface.

Prior to this dashboard, leadership teams relied on disparate Excel reports, manual email updates, and disconnected data sources across functional departments. The dashboard consolidates all of this into a single always-current view, powered by live data from SharePoint and enriched with AI-generated insights via Anthropic Claude.

Key Highlights

- 6 functional tabs covering all major business domains
- Live data pulled from SharePoint via Microsoft Graph API
- AI Leadership Summary on every tab — auto-generated insights for decision-making
- Needs Attention panel — proactively flags deals and issues requiring action
- Migration Urgency Classification — 4-level risk indicator per deal
- ETL pipeline with parallel AI processing — 58 deals processed in ~15 seconds
- Export functionality — PDF, Excel, CSV per tab
- Built on Next.js 14 / React / TypeScript / Python / Azure App Service

Project Name	Trianz Internal Leadership Dashboard
Platform	Azure App Service (B1, Linux, Central India)
Frontend	Next.js 14 / React / TypeScript / Tailwind CSS
Backend/ETL	Python 3.11 / pandas / openpyxl
AI Engine	Anthropic Claude (Haiku) via Claude API
Data Source	SharePoint (Microsoft Graph API) + Excel files
Deployment	GitLab → Azure Deployment Center (External Git)
Primary Users	Trianz Leadership, Business Heads, Functional Leads

2. Project Overview & Scope

2.1 Problem Statement

Trianz leadership teams across Finance, HR, Delivery, and Sales maintained separate Excel-based reporting systems with no unified view. Monthly/quarterly business reviews required significant manual effort to compile data from multiple teams. There was no real-time visibility into delivery health, sales pipeline status, attrition trends, or financial performance in a single place.

2.2 Solution

A web-based internal dashboard that:

- Automatically pulls data from SharePoint-hosted Excel files via Microsoft Graph API
- Processes and transforms raw Excel data into structured metrics via Python ETL scripts
- Presents all KPIs in a clean, leadership-optimized UI with drill-down capability
- Generates AI-powered summaries and action recommendations using Claude
- Flags deals and issues proactively rather than requiring manual review

2.3 Scope

Tab	Functional Area	Primary Data Source	Key Metrics
Finance	Financial Performance	Finance P&L Excel (SharePoint)	Revenue, GP, EBITDA, Margins
Sales	Sales Pipeline	SFDC Pipeline Export (SharePoint)	Pipeline, TCV, ACV, Stages
HR & People	Workforce	HR Attrition/Headcount Excel (SharePoint)	Attrition, Headcount, HC Cost
Delivery	Project Delivery	EPMO, Data Cards Excel (SharePoint)	Projects, Utilization, Margins
Active Engagements	Platform Engagements	AWS Master Billing + Deal Portfolio	ACV, QCV, Deal Status
Backlog	Platform Backlog	Forecast Tracking FY27 (SharePoint)	ACV Year, QCV Quarter

2.4 End Users

- CIO and Business Heads — executive overview via Executive tab
- Finance Leadership — Finance tab deep-dive
- HR / People Team — HR & People tab
- Delivery Leads — Delivery tab
- Sales / Platform Teams — Sales, Active Engagements, Backlog tabs

3. Technical Architecture

3.1 Tech Stack

Layer	Technology	Purpose
Frontend	Next.js 14 (App Router)	SSR/SSG page rendering
UI Framework	React 18 + TypeScript	Component library
Styling	Tailwind CSS	Utility-first CSS
Charts	Recharts	Data visualization
ETL / Data	Python 3.11 + pandas + openpyxl	Excel parsing & transformation
AI Engine	Anthropic Claude Haiku	AI summaries & recommendations
Data Layer	JSON files (local filesystem)	Intermediate data store
Data Source	SharePoint via Microsoft Graph API	Live Excel file access
Hosting	Azure App Service B1 (Linux)	Production hosting
CI/CD	GitLab → Azure Deployment Center	Automated deployment
Scheduler	node-cron	Automated ETL scheduling

3.2 System Architecture

The system is structured as a single Next.js application with an embedded Python ETL pipeline:

Data Flow:

- Step 1 — SharePoint hosts the Excel source files (Finance, HR, Delivery, Sales, Active Engagements, Backlog)
- Step 2 — Python ETL scripts connect to SharePoint via Microsoft Graph API, download files to memory, parse and transform data
- Step 3 — ETL outputs structured JSON files to the /data directory on the server
- Step 4 — Next.js pages read JSON via dataCache.ts (with in-memory caching for performance)
- Step 5 — React components render the data into KPI cards, charts, and tables
- Step 6 — Claude AI API is called per-tab to generate leadership summaries and deal recommendations

Architecture Layers:

- Presentation Layer: Next.js pages + React components + Tailwind CSS
- API Layer: Next.js API routes (/api/refresh, /api/metrics/*, /api/ai/ask)
- ETL Layer: Python scripts (etl/*.py) triggered via Node.js child_process
- Data Layer: JSON files in /data directory + in-memory cache
- Integration Layer: SharePoint (Microsoft Graph API), Claude API
- Infrastructure: Azure App Service B1, GitLab repository

3.3 Project File Structure

Path	Description
app/	Next.js App Router pages (one folder per tab)
components/	Reusable React components (KpiCard, DrillDrawer, etc.)
etl/	Python ETL scripts — one per data domain
etl/common/	Shared utilities: finder.py, SharePoint client
lib/	TypeScript utilities: dataCache, etlRunner, aiDataCache
data/	Output JSON files written by ETL (gitignored)
sample_data/	Local Excel files for development (gitignored)
public/	Static assets
startup.sh	Azure startup script (installs Python deps, starts Next.js)

4. Dashboard Features — By Tab

4.1 Finance Tab

The Finance tab provides a consolidated view of Trianz's financial performance across Services, Platforms, and Overall, covering current FY actuals, prior FY comparison, and planned targets.

KPIs & Metrics

Metric	Description	Breakdown Available
Revenue	Total revenue for the period	Services / Platforms / Overall
Direct Cost	Project-level direct costs	Services / Platforms / Overall
Project Profit	Revenue minus direct cost	Services / Platforms / Overall
Project Margin %	Project profit as % of revenue	Services / Platforms / Overall
Gross Profit	Revenue minus all delivery costs	Services / Platforms / Overall
Gross Margin %	Gross profit as % of revenue	Services / Platforms / Overall
Adjusted EBITDA	Earnings before interest, tax, depreciation	Services / Platforms / Overall
EBITDA %	EBITDA as % of revenue	Services / Platforms / Overall
SG&A Cost	Sales & marketing + G&A costs	Overall
Planned vs Actual	Current FY actuals vs planned budget	All LOBs

Filters & Views

- Period filter: Year / Quarter
- LOB filter: Overall / Services / Platforms
- Quarter breakdown: Q1, Q2, Q3, Q4 with YTD and QTD views
- Year-over-year comparison: Current FY vs Prior FY
- Planned vs Actual: Budget vs actual performance

How Data Flows

- Source: Finance P&L Excel file hosted in SharePoint (finance/ folder)
- ETL: etl/finance.py downloads the file, scans sheets by content fingerprint, parses LOB sections and metric rows dynamically — no hardcoded column/row numbers
- Output: data/finance.json with metrics nested by LOB and time period
- Frontend: app/finance/page.tsx reads the JSON and passes to FinanceCards components

4.2 HR & People Tab

The HR & People tab consolidates workforce analytics — attrition trends, headcount distribution, HC cost analysis, and open position tracking — giving leadership a complete view of people metrics.

KPIs & Metrics

Metric	Description	Breakdown Available
Attrition Rate (TTM)	Trailing 12-month attrition — voluntary and total	By BU, by segment
Monthly Attrition	Per-month attrition count and percentage	By BU, by segment
Voluntary vs Total	Split between voluntary resignation and total exits	By org group
Headcount (FTE)	Total headcount by org group	Offshore / Onsite split
HC FTE (Cost Basis)	FTE count on HC cost basis including contractors	Monthly trend
HC Cost YTD	Year-to-date headcount cost in $M	Monthly breakdown
HC Cost QTD	Quarter-to-date headcount cost	Monthly breakdown
Cost per FTE	Monthly cost divided by FTE count	Latest month
Open Positions	Total open requisitions	By criticality, by org group
Revenue Generating Reqs	Open positions that are billable/revenue-generating	Count and %

How Data Flows

- Source: 4 Excel files in SharePoint hr/ folder: Attrition, Headcount, HC Cost, Open Positions
- ETL: etl/hr.py scans each file by content fingerprint — finds period headers, subtotal rows, and column labels dynamically
- Sanity check: BU sub-totals are verified to sum to grand total (warns if >5% gap)
- Output: data/hr.json with nested attrition, headcount, hc_cost, and open_positions objects

4.3 Delivery Tab

The Delivery tab provides visibility into active project portfolio health, client coverage, resource utilization, bench status, and project-level margin performance across Services and Platforms.

KPIs & Metrics

Metric	Description	Breakdown Available
Active Clients	Unique clients with active projects	Services / Platforms
Active Projects	Total open/active project count	Services / Platforms / by sub-segment
Platforms Sub-segments	Breakdown by Concerto product line	Manage / Migrate / Modernize / DTS&DA
Utilization Rate	Billed hours as % of total hours	Overall / by Service Line
Bench Rate	Bench hours as % of total hours	Overall / by Service Line
Top 5 High Margin Projects	Highest margin projects by contribution %	Services / Platforms
Top 5 Low Margin Projects	Lowest margin projects — attention required	Services / Platforms
LARR Achieved	VMware migration ARR achieved (Direct + Partner)	Direct / Partner split

LARR Target vs Gap	Target ARR vs what has been achieved	Overall
VMs Migrated	Total VMs migrated count	Direct / Partner

How Data Flows

- Source: Multiple Excel files in SharePoint delivery/ folder — EPMO Project Admin, Data Cards (utilization), Finance Margin file, LARR Summary
- ETL: etl/delivery.py uses find_delivery_sheets() to auto-detect each file by content, then runs separate parsers for each source
- Output: data/delivery.json combining all delivery metrics into a single nested structure

4.4 Sales Tab

The Sales tab provides a pipeline view of Trianz's FY27 sales opportunities, segmented by stage, offering (Services vs Platforms), and quarter. It tracks deal counts, pipeline value, TCV, and ACV across the full sales funnel.

Pipeline Stage Buckets

Bucket	Stages Included	Description
Total Pipeline	Stage 1 to Stage 5	All active FY27 deals (excludes Stage 0 Prospects)
Advanced Pipeline	Stage 3, 4, 5	High-probability deals in solution/negotiate/booking
Initial Pipeline	Stage 1, 2	Early-stage deals in engage/develop

KPIs & Metrics

Metric	Description	Breakdown
Pipeline Annual	Total FY27 annual deal value across all stages	Services / Platforms
Pipeline Quarterly	Deal value split by Q1-Q4	Services / Platforms
Deal Count	Number of deals in each bucket	Services / Platforms
TCV (Total Contract Value)	Full contract value from Amount column	All buckets
ACV (Annual Contract Value)	Annual recurring value	All buckets
Top Deals	Top 25 deals by annual value	Services / Platforms tabs
Sub-segment Breakdown	Pipeline by Deal Classification	Assess / Manage / Migration / Modernize

How Data Flows

- Source: SFDC pipeline export Excel in SharePoint sales/ folder
- ETL: etl/sales.py auto-detects the header row (can be at row 1 or row 9), maps column aliases, filters to FY27 dates (Apr 2026 – Mar 2027)

- Stage mapping: Stage 0 (Prospect) is explicitly excluded from all pipeline buckets
- Offering classification: Deal Classification column maps Assess/Manage/Migration/Modernize → Platforms; Services deals detected separately

4.5 Active Engagements Tab

The Active Engagements tab tracks live AWS/cloud migration engagements, powered by the Deal Portfolio as the primary data source. It provides delivery teams and leadership with deal-level visibility into migration progress, blockers, customer connect status, and AI-recommended next actions.

KPIs & Metrics

Metric	Description
ACV (Year)	Annual Contract Value for active platform engagements
QCV (Quarter)	Quarterly Contract Value for the current quarter
Active Deal Count	Total deals in active/delivery status
Direct vs Partner Split	Deals by business type (Direct and Partner tabs)
Migration Urgency	4-level classification: Red / Amber / Yellow / Green
Customer Connect Flag	Flags deals with no recent customer engagement
Needs Attention Panel	Auto-flagged deals with blockers or stalled progress
AI Leadership Summary	Claude-generated summary with action table
Deal Portfolio Table	Full deal list with phase, completion %, blockers, remarks

Migration Urgency Classification

Level	Color	Meaning
Critical	Red	Deadline-driven or high-risk migration — immediate action required
High	Amber	Active driver present with near-term deadline or blocker
Medium	Yellow	Migration in progress with moderate urgency
Low	Green	No active urgency driver — monitoring phase

Needs Attention Panel

Automatically surfaces up to 5 deals that match any of these conditions:

- Active blocker recorded in the deal remarks
- Assessment or migration completion below threshold
- Deal ready to advance to next phase but not yet moved
- No customer connect recorded (customer engagement gap)

4.6 Backlog Tab

The Backlog tab tracks Platforms business backlog — the total contracted ACV (Annual Contract Value) and QCV (Quarterly Contract Value) pipeline for Concerto product lines across Direct and Partner channels.

KPIs & Metrics

Metric	Description	Breakdown
ACV (Year)	Total annual contracted backlog value	Overall / Direct / Partner
QCV (Quarter)	Quarterly contracted backlog value	Overall / Direct / Partner
By Offering	Backlog split by Concerto offering	C-Migrate / C-Manage / C-Modernize / C-Insights
By Channel	Direct vs Partner split	Per offering
AI Leadership Summary	Claude-generated backlog narrative	Auto-generated

How Data Flows

- Source: Forecast Tracking FY27 Excel in SharePoint (platforms/ folder)
- ETL: etl/platforms_backlog.py parses sheet by channel (Direct/Partner) and aggregates ACV and QCV by offering and overall
- Output: data/platforms_backlog.json

5. Data Sources & Integration

5.1 SharePoint as the Single Source of Truth

All source Excel files are hosted on a dedicated SharePoint site (Internal Dashboard site) and organized into folders by functional domain. The dashboard connects to SharePoint via the Microsoft Graph API using an Azure App Registration with client credentials flow — no user login required.

5.2 SharePoint Folder Structure

SharePoint Folder	Contains	Used By
finance/	Finance P&L Excel (current FY, prior FY, planned)	Finance tab
hr/	Attrition, Headcount, HC Cost, Open Positions Excels	HR & People tab
delivery/	EPMO Project Admin, Data Cards, Finance Margin, LARR files	Delivery tab
sales/	SFDC pipeline export Excel	Sales tab
platforms/	Forecast Tracking FY27	Backlog tab
active_engagements/	AWS Master Billing, Deal Portfolio, Forecast Tracking	Active Engagements tab

5.3 Microsoft Graph API Authentication

Authentication uses the OAuth 2.0 Client Credentials flow — the app authenticates as itself (no user required):

- Azure App Registration in the Trianz tenant
- Client ID, Client Secret, and Tenant ID stored as Azure environment variables
- Token is fetched on first request and cached in memory for the session
- SharePoint Site URL configured as an environment variable

5.4 Environment Variables Required

Variable	Purpose
SHAREPOINT_TENANT_ID	Azure AD tenant ID for authentication
SHAREPOINT_CLIENT_ID	App Registration client ID
SHAREPOINT_CLIENT_SECRET	App Registration client secret
SHAREPOINT_SITE_URL	Full SharePoint site URL (https://trianzcorp.sharepoint.com/sites/InternalDashboard)
ANTHROPIC_API_KEY	Claude API key for AI features

5.5 Fallback Mechanism

When SharePoint credentials are not configured (e.g., local development without credentials), each ETL automatically falls back to reading from the local sample_data/ folder. This means the dashboard works offline for development without requiring SharePoint access.

6. ETL Pipeline Explained

6.1 How the ETL Pipeline Works

When the Refresh button is clicked (or the scheduled job runs), the Next.js API route /api/refresh spawns Python subprocesses for each ETL function in parallel. Each ETL:

- Connects to SharePoint and downloads the relevant Excel file(s) to an in-memory buffer
- Parses and transforms the raw Excel data using openpyxl and/or pandas
- Applies business logic (metric calculations, aggregations, classifications)
- Writes a structured JSON file to the /data directory
- The Next.js data cache is cleared so the next page load reads fresh data

6.2 ETL Scripts Overview

Script	Tab	Source Files	Key Logic
etl/finance.py	Finance	Finance P&L Excel	Scans LOB headers, maps metric rows, parses Q1-Q4 + FY totals
etl/hr.py	HR & People	Attrition, Headcount, HC Cost, Open Positions	Scans period blocks, maps BU subtotals, validates sums
etl/delivery.py	Delivery	EPMO, Data Cards, Finance Margin, LARR	Multi-file parser, pandas for utilization, per-project margin scoring
etl/sales.py	Sales	SFDC Pipeline Export	Auto-detects header row, stage bucketing, FY27 date filter
etl/platforms_backlog.py	Backlog	Forecast Tracking FY27	Parses by channel (Direct/Partner), aggregates ACV/QCV by offering
etl/active_engagements.py	Active Engagements	AWS Master Billing, Forecast Tracking	Multi-sheet join, status filtering, FY27 ACV/QCV calculation
etl/deal_portfolio.py	Active Engagements	Deal Portfolio Excel	Dual-sheet (Direct/Partner) parsing, Claude AI per-deal recommendations

6.3 Robustness Design

All ETL scripts are designed to be robust against Excel file changes:

- No hardcoded row or column numbers — headers are found by scanning for label text
- Alias lists handle renamed columns (e.g., "Revenue" or "Revenue Total" both match)
- Sanity checks verify that segment totals sum to overall (warns if >3-5% gap)
- Graceful fallback — if an optional section is missing (e.g., LARR file), the ETL continues without it

6.4 AI Integration in ETL

The Deal Portfolio ETL (etl/deal_portfolio.py) integrates Claude Haiku for per-deal AI recommendations:

- For each deal, Claude is given the deal context (client, phase, blockers, remarks, migration reason, deadline)
- Claude returns structured JSON: recommended_next_step, urgency_level, action_category
- ThreadPoolExecutor parallelizes all Claude API calls — 58 deals processed in ~15 seconds
- Results are embedded in the deal_portfolio.json output

6.5 Scheduled Refresh

The dashboard includes a node-cron scheduler that automatically runs ETL jobs on a weekly schedule. Finance, Sales, HR, and Delivery are scheduled every Monday at 08:00 IST. Active Engagements and Backlog also run on startup to ensure data is available immediately after deployment.

7. Advanced Features

7.1 AI Leadership Summary

Every tab includes an AI Leadership Summary panel powered by Anthropic Claude. The summary is:

- Auto-generated based on the live data currently loaded in the dashboard
- Structured with a clear heading, sub-points, and an action table
- The action table categorizes deals/items into: Drive Closure, Get Clarity on Rationale & Timeline, Customer Connect Missing
- Includes decision-maker extraction and deadline/driver context for each deal
- Collapsible — leadership can expand/collapse as needed
- Labelled "Auto-generated · Based on live data" so users know it is AI-generated

7.2 Needs Attention Panel

The Needs Attention panel proactively surfaces the top 5 deals or items requiring leadership attention, without requiring the user to manually scan all data. It flags:

- Deals with active blockers recorded in remarks
- Deals with low assessment or migration completion percentages
- Deals ready to advance but stalled in current phase
- Deals with no customer connect recorded in the engagement tracker

7.3 Drill-Down Drawers & Modals

Every KPI card on the dashboard is clickable and opens a Drill Drawer panel with the full breakdown behind that metric. For example:

- Clicking "Headcount (FTE)" opens a drawer showing headcount by org group with offshore/onsite split
- Clicking "Open Positions" shows breakdown by criticality level and org group
- Clicking "HC Cost YTD" shows monthly cost breakdown, QTD, YTD, and cost per FTE
- Deal Portfolio rows are expandable to show full remarks, blockers, and AI recommendations
- Phase Chart modals show assessment and migration completion by phase

7.4 Export Functionality

Each tab includes a Download PDF button that exports the current tab view to a formatted PDF document. The export captures:

- All KPI cards and their current values
- Charts and visualizations

- Data tables in the current filtered state
- The AI Leadership Summary if expanded

7.5 Migration Urgency Classification

The Active Engagements tab includes a 4-level Migration Urgency badge per deal, derived from the reason_for_migration and customer_deadlines fields:

- Red (Critical) — Deadline-driven migration with imminent risk
- Amber (High) — Active migration driver with near-term deadline or significant blocker
- Yellow (Medium) — Migration in progress, moderate urgency
- Green (Low) — No active urgency driver, monitoring phase

The badge is expandable to show the full migration driver text and deadline context inline in the deal table.

7.6 Period & Offering Filters

Most tabs include page-level filters that control the entire tab view:

- Period filter: Year (FY total) / Quarter (QTD view)
- Offering filter: Overall / Services / Platforms
- Filters persist within the session and update all components on the page simultaneously

8. Data & Refresh Operations

8.1 How the Refresh Button Works

The Refresh Now button in the top navigation bar triggers a full data refresh for all tabs:

Step	What Happens
1. API Call	Browser sends POST request to /api/refresh endpoint
2. ETL Spawn	Next.js spawns Python subprocesses for each ETL function in parallel
3. SharePoint Download	Each ETL authenticates to SharePoint and downloads its Excel file(s)
4. Data Processing	Excel files are parsed, transformed, and metrics calculated
5. AI Processing	Claude API called for deal recommendations (Deal Portfolio ETL only)
6. JSON Written	Each ETL writes its output to /data/*.json on the server
7. Cache Cleared	Next.js in-memory data cache is cleared
8. UI Updates	Refresh Details dropdown shows success/failure per ETL with duration
9. Page Reload	Next page visit reads fresh data from the updated JSON files

8.2 Refresh Status Indicators

The top navigation bar shows the last refresh timestamp and a status badge:

- Green "Refresh Now" button — ready to refresh
- "LAST REFRESH: X ago" — shows how long since last successful refresh
- "⚠ N/4" badge — shows how many ETLs failed in the last refresh
- Clicking the badge expands Refresh Details showing each ETL's status and duration
- "X Failed" or "✓ Done" shown per ETL — with duration in seconds

8.3 Tab-Specific Refresh

Each tab also has its own individual refresh button (e.g., "Refresh Backlog", "Refresh Active Engagements") that triggers only that tab's ETL, allowing targeted updates without running all ETLs.

8.4 Data Sources Per Tab Summary

Tab	SharePoint Folder	Key Files
Finance	finance/	Finance Data_P&L and Plan.xlsx

HR & People	hr/	Attrition.xlsx, HR Data - Headcount & Open Positions.xlsx, FTE Cost YTD FY27, Metrics for Executive Dashboard.xlsx
Delivery	delivery/	EPMO_Project_Admin.xlsx, For data cards - revenue & HC.xlsx, Finance Revenue and Margin data FY27.xlsx, LARR Summary file
Sales	sales/	Raw data (SFDC pipeline export).xlsx
Active Engagements	active_engagements/	AWS Master Billing.xlsx, Deal Portfolio.xlsx, Forecast Tracking FY 27.xlsx
Backlog	platforms/	Forecast Tracking FY 27.xlsx

8.5 Refresh Frequency & Scheduling

- Manual refresh: Anytime via the Refresh Now button
- Scheduled refresh: Finance, Sales, HR, Delivery — every Monday at 08:00 IST (auto-scheduled via node-cron)
- Startup refresh: Active Engagements and Backlog auto-refresh on every server startup
- Data persists in /data folder — survives server restarts (stored under /home on Azure)

9. Data Dictionary & Schema

9.1 Finance Metrics

Field	Definition	Unit
revenue	Total recognized revenue for the period	$M
direct_cost	Direct project delivery costs	$M
project_profit	Revenue minus direct cost	$M
project_margin_pct	Project profit divided by revenue	%
gross_profit	Revenue minus all delivery costs including unbilled	$M
gross_margin_pct	Gross profit as % of revenue	%
adjusted_ebitda_value	EBITDA adjusted for non-recurring items	$M
adjusted_ebitda_pct	Adjusted EBITDA as % of revenue	%
ytd	Year-to-date (full FY total)	$M
qtd	Quarter-to-date (latest quarter)	$M
prev_fy	Prior fiscal year total	$M

9.2 HR Metrics

Field	Definition	Unit
voluntary_count	Number of voluntary resignations	Count
total_count	Total attrition (voluntary + involuntary)	Count
voluntary_pct	Voluntary attrition as % of headcount	%
total_pct	Total attrition as % of headcount	%
ttm	Trailing 12 months (Apr to Mar)	Period key
headcount.total	Total active employee count	Count
headcount.offshore	Offshore employee count	Count
headcount.onsite	Onsite employee count	Count
hc_cost.ytd	Year-to-date headcount cost	$
hc_cost.qtd	Quarter-to-date headcount cost	$
open_positions.total	Total open requisitions	Count
by_criticality	Open positions grouped by priority level	Count per level

9.3 Sales & Pipeline Metrics

Field	Definition	Unit
annual	Sum of Q1+Q2+Q3+Q4 deal values	$
deal_count	Number of deals in bucket	Count
tcv.total	Total Contract Value (Amount column)	$
acv.total	Annual Contract Value (ACV FY27 column)	$
pipeline_stages_1_to_5	All FY27 deals in Stages 1-5	Bucket

advanced_pipeline_stages_3_to_5	High-probability deals	Bucket
initial_pipeline_stages_1_to_2	Early-stage deals	Bucket

10. User Guide & Walkthrough

10.1 Navigating the Dashboard

The top navigation bar contains tab links for all sections: Executive, Finance, Sales, Backlog, Active Engagements, HR & People, and Delivery. Click any tab to navigate to it. The current tab is highlighted with an underline.

10.2 Using Filters

- Period filter (Year / Quarter): Changes whether KPIs show FY total or QTD values — appears at the top of most tabs
- Offering filter (Overall / Services / Platforms): Segments all metrics by business line — appears as a pill selector
- Filters apply to all components on the current tab simultaneously

10.3 Interacting with KPI Cards

- Every KPI card is clickable — click to open the Drill Drawer with the full breakdown
- Hover over a card to see the hover highlight (teal ring)
- Cards showing "Pending" mean data hasn't been loaded yet — click Refresh Now
- Cards showing a value with a sub-label (e.g., "Offshore 800 · Onsite 200") provide quick context without needing to drill down

10.4 Using the AI Leadership Summary

- Click "expand" on the AI Leadership Summary panel to see the full AI-generated narrative
- The summary includes a structured action table — use this to identify which deals need attention
- The summary updates automatically when data is refreshed
- It is clearly labelled as AI-generated — always verify against source data for critical decisions

10.5 Refreshing Data

- Click "Refresh Now" in the top-right navigation to trigger a full data refresh
- The refresh runs all ETLs in parallel — typically takes 30-90 seconds
- Watch the ⚠ badge — it shows how many ETLs succeeded/failed
- Click the badge to see per-ETL status and duration
- Individual tabs also have their own refresh button for targeted updates

10.6 Downloading Reports

- Click "Download PDF" on any tab to export the current view as a PDF
- The PDF captures KPIs, charts, and tables in the current filter state
- Useful for sharing with stakeholders who don't have dashboard access

11. Integration Points

System	Integration Type	Purpose	Direction
SharePoint (InternalDashboard site)	Microsoft Graph API (Client Credentials)	Download source Excel files	Read from SharePoint
Anthropic Claude API	REST API (claude-haiku model)	AI summaries, deal recommendations, next steps	Read (API call)
Azure App Service	Git-based deployment (Oryx build)	Hosting the Next.js application	Deployment target
GitLab Repository	External Git (Azure Deployment Center)	Source code version control and deployment trigger	Source of truth
node-cron	Internal scheduler	Scheduled ETL refresh jobs	Internal trigger

Note: The dashboard does not write back to any external system. It is a read-only consumer of SharePoint data and a read-only caller of the Claude API. All data transformations happen internally.

12. Assumptions & Constraints

12.1 Data Assumptions

- All source Excel files follow the format established during ETL development — major structural changes (e.g., renamed LOB headers, removed columns) require ETL updates
- Finance ETL assumes Services + Platforms ≈ Overall (sanity check threshold: 3%)
- HR ETL assumes BU subtotals sum to grand total (sanity check threshold: 5%)
- Sales ETL assumes all deals with Deal Classification as Assess/Manage/Migration/Modernize are Platforms deals
- FY27 date range is April 1, 2026 – March 31, 2027 — deals outside this range are excluded from pipeline calculations

12.2 Known Limitations

- The dashboard does not support real-time streaming data — refresh is manual or scheduled
- AI Leadership Summaries are generated at refresh time and may not reflect very recent deal activity
- PDF export captures the current browser view — very large tables may be truncated
- Data is only as current as the last ETL run — the "Last Refresh" indicator shows when data was last updated
- Python ETL requires packages (pandas, openpyxl, anthropic, requests) to be installed on the server — on Azure Node.js image this requires additional setup

12.3 What Is Not Covered

- Write-back to SharePoint or Salesforce — the dashboard is read-only
- Real-time deal updates — data refreshes on a schedule, not on every deal change
- User-level personalization — all authenticated users see the same data
- Historical trend analysis beyond what is in the source Excel files

13. Performance & Reliability

Metric	Expected Value	Notes
Page Load Time	< 2 seconds	Data served from JSON cache — no DB query on page load
Full Refresh Duration	30–90 seconds	Depends on SharePoint download speed and Claude API latency
Deal Portfolio AI Processing	~15 seconds for 58 deals	Parallelized with ThreadPoolExecutor
Data Staleness	Up to 7 days (scheduled) or manual	Weekly auto-refresh + on-demand manual refresh
Uptime	Azure B1 SLA: 99.95%	Standard Azure App Service SLA
Concurrent Users	Supports leadership team usage	No heavy concurrent load expected

The dashboard uses Next.js in-memory caching (dataCache.ts) to serve tab data without hitting the filesystem on every request. The cache is cleared only when an ETL refresh completes, ensuring fast page loads between refreshes.

14. FAQ & Troubleshooting

14.1 Frequently Asked Questions

Q: Why is a tab showing "Pending" for all metrics?

A: The ETL for that tab has not run yet, or the last run failed. Click "Refresh Now" to trigger a fresh data load. If it still shows Pending after refresh, check the ⚠ badge for error details.

Q: Why does the AI Leadership Summary say "Data unavailable"?

A: The AI summary requires data to be present in the tab. If the underlying metrics show Pending, refresh data first. If data is present but summary still fails, it may be a Claude API connectivity issue.

Q: How often should I refresh?

A: Finance, Sales, HR, and Delivery are auto-refreshed every Monday at 08:00 IST. For the most current data before an important meeting, click Refresh Now manually. Active Engagements and Backlog also refresh on server startup.

Q: Can I see data from a specific month?

A: Finance tab supports Year/Quarter period filters. Monthly breakdowns are available in the Drill Drawer by clicking individual KPI cards. Full month-by-month data is available in the source Excel files.

Q: The refresh shows "X Failed" — what does that mean?

A: One or more ETL scripts encountered an error. Click the ⚠ badge to see which ETL failed and the error message. Common causes: SharePoint file not found, Excel format changed, Claude API timeout.

14.2 Common Issues & Fixes

Issue	Likely Cause	Fix
Tab shows all Pending after refresh	ETL failed to write JSON	Check ⚠ badge for ETL error details
AI Summary not generating	ANTHROPIC_API_KEY not set or invalid	Verify env var is set in Azure Configuration
SharePoint ETL failing	Credentials expired or permissions changed	Verify SharePoint env vars in Azure Configuration
Excel format changed	Source team renamed a column or sheet	Update alias list in the relevant ETL script
Refresh takes very long	SharePoint slow or Claude API latency	Wait for completion; consider running individual tab refresh
Download PDF shows empty	Data not loaded in current tab	Refresh data first, then export

15. Appendix

15.1 ETL File Reference

File	Input	Output
etl/finance.py	SharePoint: finance/ folder	data/finance.json
etl/hr.py	SharePoint: hr/ folder (4 files)	data/hr.json
etl/delivery.py	SharePoint: delivery/ folder	data/delivery.json
etl/sales.py	SharePoint: sales/ folder	data/sales.json
etl/platforms_backlog.py	SharePoint: platforms/ folder	data/platforms_backlog.json
etl/active_engagements.py	SharePoint: active_engagements/ folder	data/active_engagements.json
etl/deal_portfolio.py	SharePoint: active_engagements/ folder	data/deal_portfolio.json
etl/sharepoint_client.py	N/A (shared utility)	N/A — provides download_folder_to_dir(), download_to_bytes()
etl/common/finder.py	N/A (shared utility)	N/A — provides file/sheet discovery functions

15.2 API Routes Reference

Route	Method	Purpose
/api/refresh	POST	Trigger ETL refresh for all or specific functions
/api/refresh	GET	Get last refresh log and history
/api/metrics/finance	GET	Get Finance tab data (reads from JSON cache)
/api/metrics/hr	GET	Get HR tab data
/api/metrics/delivery	GET	Get Delivery tab data
/api/metrics/sales	GET	Get Sales tab data
/api/ai/ask	POST	Ask Claude a question about dashboard data
/api/schedule	GET/POST	View or manage scheduled ETL jobs

15.3 Glossary

Term	Definition
ACV	Annual Contract Value — recurring annual revenue from a deal
QCV	Quarterly Contract Value — ACV for the current quarter
TCV	Total Contract Value — full lifetime value of a contract
ETL	Extract, Transform, Load — the data pipeline that pulls from Excel and produces JSON
LOB	Line of Business — Services, Platforms, or Overall
FTE	Full-Time Equivalent — headcount unit including contractors on cost basis
TTM	Trailing Twelve Months — rolling 12-month period

LARR	Licensed ARR — Annual Recurring Revenue from licensed VMware migration deals
ARR	Annual Recurring Revenue
EPMO	Enterprise Project Management Office — maintains the project admin tracker
SharePoint	Microsoft SharePoint — document management platform used as data source
Graph API	Microsoft Graph API — REST API for accessing SharePoint and other M365 services
Oryx	Azure's open-source build system used to build Node.js apps on App Service
Claude/Haiku	Anthropic's AI model used for generating leadership summaries and deal recommendations

15.4 Key Contacts

Role	Name	Area
Project Lead / Developer	Deepak V	Internal Dashboard — all tabs
Stakeholder / Reviewer	Priyesh	Project oversight, requirements
Portal Integration	Prabhat	Production portal deployment
Finance Data Owner	Finance Team (Shreyansh)	Finance P&L Excel maintenance
HR Data Owner	Ayushi	Attrition, Headcount, HC Cost data
Delivery Data Owner	Shobha	EPMO, Delivery metrics
IT / SharePoint	Saravanavignesh	SharePoint app registration, IT support

End of Document — Trianz Internal Leadership Dashboard v1.0